Radiation-Hydrodynamics Simulation

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1 Hydrodynamics scheme

The scheme uses the principle of operator splitting, where the hydrodynamics and radiation are solved separately. The hydrodynamics is solved using a finite volume Lagrangian method (e.g. Richtmyer-Morton) and the radiation is solved using a finite difference method. The coupling between the two is done through the source terms in the hydrodynamics equations, which are updated after each radiation step.

1.1 Hydrodynamics step

The following uses the Euler equations:

$$\frac{\partial u}{\partial t} = -\frac{\partial P}{\partial m} \tag{1}$$

$$\frac{\partial V}{\partial t} = \frac{\partial u}{\partial m} \tag{2}$$

$$\frac{\partial e}{\partial t} = -P\frac{\partial u}{\partial m} \tag{3}$$

$$p = r\rho e \tag{4}$$

$$P = p + q \tag{5}$$

where E_{rad} is the energy exchange with the radiation field, which is updated after the radiation step (kept aside for now). In the lagrangian scheme, we begin with a certain pressure that applies a force on the mass elements, which causes them to accelerate. The scheme goes as follows:

1. Compute acceleration from the pressure gradient:

$$a_i^n = -2 \cdot \frac{\left(p^n_{i+\frac{1}{2}} + q^n_{i+\frac{1}{2}}\right) - \left(p^n_{i-\frac{1}{2}} + q^n_{i-\frac{1}{2}}\right)}{m_{i+\frac{1}{2}} + m_{i-\frac{1}{2}}} \tag{6}$$

2. Update velocity using the acceleration:

$$u_i^{n+\frac{1}{2}} = u_i^n + a_i^n \Delta t \tag{7}$$

3. Update node positions using the updated velocities (which is exactly a helper step for updating the volume):

$$x_i^{n+1} = x_i^n + u_i^{n+\frac{1}{2}} \Delta t \tag{8}$$

4. Update the volume and density of each cell using the new node positions:

$$V_{i+\frac{1}{2}}^{n+1} = \frac{x_{i+1}^{n+1} - x_i^{n+1}}{m_{i+\frac{1}{2}}} \tag{9}$$

$$\rho_{i+\frac{1}{2}}^{n+1} = \frac{1}{V_{i+\frac{1}{2}}^{n+1}} \tag{10}$$

5. Calculate artificial viscosity q to handle shocks, using the von Neumann-Richtmyer prescription:

$$q_{i+\frac{1}{2}}^{n+1} = \begin{cases} \sigma \rho_{i+\frac{1}{2}}^{n+1} \left(u_{i+1}^{n+1} - u_i^{n+1}\right)^2 & \text{if } u_i^{n+1} > u_{i+1}^{n+1} \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

6. Calculate the new spcific internal energy using the updated pressure and velocity:

$$e_{i+\frac{1}{2}}^{n+1} = \frac{e_{i+\frac{1}{2}}^n - \frac{1}{2}\left(p_{i+\frac{1}{2}}^n + q_{i+\frac{1}{2}}^n + q_{i+\frac{1}{2}}^{n+1}\right)\left(\frac{V_{i+\frac{1}{2}}^{n+1} - V_{i+\frac{1}{2}}^n}{m_{i+\frac{1}{2}}}\right)}{1 + \frac{\gamma-1}{2}\rho_{i+\frac{1}{2}}^{n+1}\frac{V_{i+\frac{1}{2}}^{n+1} - V_{i+\frac{1}{2}}^n}{m_{i+\frac{1}{2}}}} \tag{12}$$

7. Calculate the new pressure using the equation of state:

$$p_{i+\frac{1}{2}}^{n+1} = (\gamma - 1)\rho_{i+\frac{1}{2}}^{n+1} e_{i+\frac{1}{2}}^{n+1} \tag{13}$$

8. Compute acceleration from the **new** pressure gradient:

$$a_i^{n+1} = -2 \cdot \frac{\left(p_{i+\frac{1}{2}}^{n+1} + q_{i+\frac{1}{2}}^{n+1}\right) - \left(p_{i-\frac{1}{2}}^{n+1} + q_{i-\frac{1}{2}}^{n+1}\right)}{m_{i+\frac{1}{2}} + m_{i-\frac{1}{2}}} \tag{14}$$

9. Update velocity using the new acceleration:

$$u_i^{n+1} = u_i^{n+\frac{1}{2}} + a_i^{n+1}\frac{\Delta t}{2} \tag{15}$$

and go back to step 2 for the next iteration. Make sure to understand the reason for the half time step in the velocity update - it makes the scheme second-order and less noisy.

1.2 Coupling with radiation

This is all good for the hydrodynamics step, but we need to couple it with radiation! After step 9, we define:

$$e^* = e^{n+1} \tag{16}$$

which means that the energy isn't updated yet! we then assume that $U_m^n = \rho e^*$ is the energy density of the matter.

2 Radiation Scheme

2.1 Physical PDEs

1. Writing the diffusion equation for the radiation energy E:

$$\frac{\partial E}{\partial t} + \rho\frac{\partial F}{\partial m} = \chi c\sigma(U_R - E) \tag{17}$$

$$\frac{\partial U_m}{\partial t} = -\chi c\sigma(U_R - E) \Rightarrow \frac{1}{\beta}\frac{\partial U_R}{\partial t} = -\chi c\sigma(U_R - E) \tag{18}$$

$$F = -D\rho\frac{\partial E}{\partial m} \quad \left(D = \frac{c}{3\sigma(T,\rho)}\right) \tag{19}$$

In the equation here, one assumes that during the radiation substep we keep ρ^* (to be defined later) fixed, and then linearize $U_R(U_m)$ locally around U_m^* to get the relation $\frac{1}{\beta}\frac{\partial U_R}{\partial t} = -\chi c\sigma(U_R - E)$, where $\beta = \frac{dU_R}{dU_m}|_{U_m^*}$ is the coupling coefficient between the radiation and matter that satifies $dU_R \approx \beta dU_m$ around U_m^*.

2. The self similar relations for the opacity (and thus the diffusion constant):

$$\frac{1}{\kappa_R} = gT^\alpha\rho^{-\lambda} \tag{20}$$

$$\frac{1}{\sigma(T,\rho)} = \frac{1}{\rho\kappa_R(T)} = gT^{\alpha}\rho^{-\lambda-1} \Rightarrow D = \frac{c}{3}gT^{\alpha}\rho^{-\lambda-1} \tag{21}$$

and the self similar relation for the internal energy:

$$e = fT^{\gamma}\rho^{-\mu} \tag{22}$$

where e is the specific internal energy, which means the material energy density is:

$$U_m(T) = \rho e = fT^{\gamma}\rho^{-\mu+1} \tag{23}$$

3. Notice that we've converted the space derivatives to mass derivatives, using the relation:

$$\frac{\partial}{\partial x} = \frac{\partial m}{\partial x}\frac{\partial}{\partial m} = \rho\frac{\partial}{\partial m} \tag{24}$$

as done in the Appendix.

2.2 Finite Difference Scheme

1. One should notice that the flux is defined at the nodes, while the energy is defined at the cell centers. The definition for the flux is given by:

$$F_i^n = -\rho_{i+\frac{1}{2}}^n D_i^n \frac{E_{i+\frac{1}{2}}^n - E_{i-\frac{1}{2}}^n}{\Delta m_i} \tag{25}$$

Where I used here the notation of:

$$\Delta m_i := \frac{1}{2}\left((m_{i+1} - m_i) + (m_i - m_{i-1})\right) = \frac{1}{2}(\Delta m_{i+\frac{1}{2}} + \Delta m_{i-\frac{1}{2}}) \tag{26}$$

where the lagrangina coordinate m_i for $i = 0, ..., N$ is defined as the mass of the material up to the vertex located at x_i. Therefore $m_i = \sum_{j=0}^{i} \Delta m_{i+\frac{1}{2}}$. where $\Delta m_{i+\frac{1}{2}} = m_{i+1} - m_i$ is the mass of the cell $j = i + \frac{1}{2}$ $(j = 1, ..., N)$ between the vertices i and $i+1$. It also be denoted $\Delta m_i = m_{i+\frac{1}{2}} - m_{i-\frac{1}{2}}$ where $m_{i+\frac{1}{2}} = m_i + \frac{1}{2}\Delta m_{i+\frac{1}{2}}$.

$$\Delta m_i = m_i + \frac{1}{2}\Delta m_{i+\frac{1}{2}} - \left(m_{i-1} + \frac{1}{2}\Delta m_{i-\frac{1}{2}}\right) \tag{27}$$

NOTE that in our case $\Delta m_{i+\frac{1}{2}}$ is constant for all i and therefore $\Delta m_i = const$ and is a **predefined parameter** of the simulation.

2. The diffusion constant (which is also defined at the nodes) is an harmonic mean of the diffusion constants at the surrounding cell centers:

$$D_i^n = \frac{2D_{i+\frac{1}{2}}^n D_{i-\frac{1}{2}}^n}{D_{i+\frac{1}{2}}^n + D_{i-\frac{1}{2}}^n} \tag{28}$$

I need to ask Yuval where he implemented simple average instead.

3. We've seen that one can **eliminate the effective radiation density** U_R from the finite-difference equation for the matter energy. The discretized equation for the material energy density is:

$$\frac{1}{\beta_{i+\frac{1}{2}}^n} \frac{[U_R]_{i+\frac{1}{2}}^{n+1} - [U_R]_{i+\frac{1}{2}}^n}{\Delta t} = \chi c \sigma_{i+\frac{1}{2}}^n (E_{i+\frac{1}{2}}^{n+1} - [U_R]_{i+\frac{1}{2}}^{n+1}) \tag{29}$$

$$(1 + \beta_{i+\frac{1}{2}}^n \Delta t \chi c \sigma_{i+\frac{1}{2}}^n)[U_R]_{i+\frac{1}{2}}^{n+1} = \beta_{i+\frac{1}{2}}^n \Delta t \chi c \sigma_{i+\frac{1}{2}}^n E_{i+\frac{1}{2}}^{n+1} + [U_R]_{i+\frac{1}{2}}^n \tag{30}$$

We want to solve for $[U_R]_{i+\frac{1}{2}}^{n+1}$. First, we can denote $A_{i+\frac{1}{2}}^n = \beta_{i+\frac{1}{2}}^n \Delta t \chi c \sigma_{i+\frac{1}{2}}^n$ as the coupling coefficient between the radiation and matter, which gives us:

$$(1 + A_{i+\frac{1}{2}}^n)[U_R]_{i+\frac{1}{2}}^{n+1} = A_{i+\frac{1}{2}}^n E_{i+\frac{1}{2}}^{n+1} + [U_R]_{i+\frac{1}{2}}^n \tag{31}$$

$$\boxed{[U_R]_{i+\frac{1}{2}}^{n+1} = \frac{A_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} E_{i+\frac{1}{2}}^{n+1} + \frac{1}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n} \tag{32}$$

4. Furthermore, after subtituting the expressions for $[U_R]_{i+\frac{1}{2}}^{n+1}$ & F_i^n into the discretized equation for the radiation energy E, we get a tridiagonal implicit scheme for E_i^{n+1}:

$$\boxed{a_i^n E_{i-\frac{1}{2}}^{n+1} + b_i^n E_{i+\frac{1}{2}}^n + c_i^n E_{i+\frac{3}{2}}^{n+1} = d_i^n} \tag{33}$$

where the coefficients $a_i^n, b_i^n, c_i^n, d_i^n$ are calculated in the appendix.

5. We notice that the coefficients $a_i^n, b_i^n, c_i^n, d_i^n$ depend on the hydrodynamics variables (e.g. density, temperature) through the diffusion constant D and the opacity σ. The dependency is given through Rosen's formulae for the opacity and the internal energy. First, one should find the temperature at the previous time step:

$$T_{i+\frac{1}{2}}^n = \sqrt[4]{\frac{[U_R]_{i+\frac{1}{2}}^n}{a}} \tag{34}$$

and then substitute it into the relations for $D(T), \beta(T)$:

$$D^n_{i+\frac{1}{2}} = \frac{c}{3} g \left[T^n_{i+\frac{1}{2}}\right]^{\alpha} \left[\rho^n_{i+\frac{1}{2}}\right]^{-\lambda-1} \tag{35}$$

$$\beta^n_{i+\frac{1}{2}} = \frac{dU_R}{dU_m} = \frac{4a}{f\gamma} \left[T^n_{i+\frac{1}{2}}\right]^{4-\gamma} \left[\rho^n_{i+\frac{1}{2}}\right]^{\mu-1} \tag{36}$$

$$\frac{1}{\sigma^n_{i+\frac{1}{2}}} = \frac{1}{\rho\kappa_R(T)} = g \left[T^n_{i+\frac{1}{2}}\right]^{\alpha} \left[\rho^n_{i+\frac{1}{2}}\right]^{-\lambda-1} \tag{37}$$

NOTE: In the coupled scheme, we will use the temperature $T^*_{i+\frac{1}{2}}$ computed from the hydrodynamics step **directly** to calculate D, β, σ and substitute in the tridiagonal system, no need to go through U_R in the coupled scheme.

6. To sum up, the complete algorithm for the radiation scheme is as follows:

   (a) Compute the temperature at the previous time step using the effective radiation density U_R.

   (b) Compute the diffusion constant D, the coupling coefficient β and the opacity σ using the temperature and density at the previous time step.

   (c) Compute the coefficients $a^n_i, b^n_i, c^n_i, d^n_i$ for the tridiagonal system.

   (d) Solve the tridiagonal system for E^{n+1}_i using the Thomas algorithm.

   (e) Update the effective radiation density U^{n+1}_R using the expression derived above.

3 Coupling the two schemes

1. We first perform the hydrodynamics step, which gives us the updated density ρ^{n+1} and specific internal energy e^{n+1}. We call it e^* because we haven't updated the energy with the radiation source term yet.

2. In the radiation scheme, the self-similar profiles for the opacity and internal energy depend on the material temperature. Since we know that $U_m = \rho e^*$, we can compute the temperature at the previous time step using the relation:

$$e^*_{i+\frac{1}{2}} = f \left[T^*_{i+\frac{1}{2}}\right]^{\gamma} \left[\rho^*_{i+\frac{1}{2}}\right]^{-\mu} \Rightarrow T^*_{i+\frac{1}{2}} = \left(\frac{e^*_{i+\frac{1}{2}}}{f} \left[\rho^*_{i+\frac{1}{2}}\right]^{\mu}\right)^{\frac{1}{\gamma}} \tag{38}$$

3. Apply the radiation scheme using the temperature and density at the previous time step to compute β, D and σ which give us the coefficients for the tridiagonal system, which we solve to get $E^{n+1}_{i+\frac{1}{2}}$ and $[U_R]^{n+1}_{i+\frac{1}{2}}$.
NOTE: The chat proposes to perform a Picard loop, where we update the temperature using the radiation energy until convergence.

4. We then update the specific internal energy after the second source term of radiation is applied:

$$e^{n+1}_{i+\frac{1}{2}} = \frac{[U_m]^{n+1}_{i+\frac{1}{2}}}{\rho^{n+1}_{i+\frac{1}{2}}} \tag{39}$$

5. Finally, we update the hydrodynamic pressure using the equation of state:

$$p^{n+1}_{i+\frac{1}{2}} = r\rho^{n+1}_{i+\frac{1}{2}} e^{n+1}_{i+\frac{1}{2}} \tag{40}$$

and go back to the hydrodynamics step for the next iteration!

4 Corrections

4.1 Correcting the scheme with black physics

1. In the radiation diffusion equation, the finite-difference form that we reached previously is:

$$LHS = \frac{\partial E}{\partial t} + \rho \frac{\partial F}{\partial m} \tag{41}$$

$$\Updownarrow \tag{42}$$

$$LHS = \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n}_{i+\frac{1}{2}}}{\Delta t} - \frac{\rho^*_{i+\frac{1}{2}}}{m_{i+\frac{1}{2}}} \overbrace{\left(\rho^n_{i+\frac{3}{2}} \frac{2D^n_{i+\frac{1}{2}} D^n_{i+\frac{3}{2}}}{D^n_{i+\frac{1}{2}} + D^n_{i+\frac{3}{2}}} \frac{E^{n+1}_{i+\frac{3}{2}} - E^{n+1}_{i+\frac{1}{2}}}{m_{i+\frac{3}{2}}} \right)}^{-F^{n+1}_{i+1}}$$
$$+ \frac{\rho^*_{i+\frac{1}{2}}}{m_{i+\frac{1}{2}}} \overbrace{\left(\rho^n_{i-\frac{1}{2}} \frac{2D^n_{i-\frac{1}{2}} D^n_{i+\frac{1}{2}}}{D^n_{i-\frac{1}{2}} + D^n_{i+\frac{1}{2}}} \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n+1}_{i-\frac{1}{2}}}{m_{i+\frac{1}{2}}} \right)}^{-F^{n+1}_{i}} \tag{43}$$

but wait! If we assume black physics, the radiation energy should be changed to E^* and not E^{n+1}. That's why the new LHS is:

$$LHS = \frac{E^*_{i+\frac{1}{2}} - E^n_{i+\frac{1}{2}}}{\Delta t} - \frac{\rho^*_{i+\frac{1}{2}}}{m_{i+\frac{1}{2}}} \overbrace{\left(\rho^n_{i+\frac{3}{2}} \frac{2D^n_{i+\frac{1}{2}} D^n_{i+\frac{3}{2}}}{D^n_{i+\frac{1}{2}} + D^n_{i+\frac{3}{2}}} \frac{E^{n+1}_{i+\frac{3}{2}} - E^{n+1}_{i+\frac{1}{2}}}{m_{i+\frac{3}{2}}} \right)}^{-F^{n+1}_{i+1}}$$
$$+ \frac{\rho^*_{i+\frac{1}{2}}}{m_{i+\frac{1}{2}}} \overbrace{\left(\rho^n_{i-\frac{1}{2}} \frac{2D^n_{i-\frac{1}{2}} D^n_{i+\frac{1}{2}}}{D^n_{i-\frac{1}{2}} + D^n_{i+\frac{1}{2}}} \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n+1}_{i-\frac{1}{2}}}{m_{i+\frac{1}{2}}} \right)}^{-F^{n+1}_{i}} \tag{44}$$

2. And the RHS remains implicit, so it's the same as before:

$$RHS = \chi c \sigma (U_R - E) \tag{45}$$

$$\Updownarrow \tag{46}$$

$$\begin{aligned} RHS &= \chi c \sigma^n_{i+\frac{1}{2}} \left(\frac{A^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}} + \frac{1}{1 + A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - E^{n+1}_{i+\frac{1}{2}} \right) \\ &= \chi c \sigma^n_{i+\frac{1}{2}} \left(\frac{1}{1 + A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - \frac{1}{1 + A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}} \right) \\ &= \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}} \end{aligned} \tag{47}$$

3. Conclusion: The new LHS and RHS define a new tridiagonal system of equations for the radiation energy density E^*, which creates a new sparse matrix that needs to be solved. Ask Shay about this conclusion - if it's correct, it requires a new algebric derivation of the tridiagonal constants, and implementation in python.

4. CORRECTION: My conclusion is incorrect. E^n should be replaced by E^* and NOT E^{n+1}. It makes sense because we suppose that the radiation energy should also be changed.

5. Implementation: The only coefficient that had previously (prior to the correction) included E^n is d^n. The new formula for d^n is:

$$d_i^n = \frac{\chi c \sigma_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n + \frac{1}{\Delta t} E_{i+\frac{1}{2}}^* \tag{48}$$

or in language of arrays:

$$d_j^n = \frac{\chi c \sigma_j^n}{1 + A_j^n} [U_R]_j^n + \frac{1}{\Delta t} E_j^* \tag{49}$$

4.2 Black (LTE) radiation scheme

1. One can also solve the rad-hydro scheme under the LTE assumption that leads to the following simpler equation:

$$\begin{aligned} \frac{\partial E}{\partial t} + \frac{\partial F}{\partial x} &= Q \\ C_V \frac{\partial T}{\partial t} &= -Q \end{aligned} \tag{50}$$

where we denoted:

$$\begin{aligned} Q &= \Sigma_a' (a T_m^4 - E) \\ F &= -D \frac{\partial E}{\partial x} \end{aligned} \tag{51}$$

$$\Downarrow \tag{52}$$

$$\begin{aligned} \frac{\partial e}{\partial t} + \frac{\partial F}{\partial x} &= 0 \\ F &= -D \frac{\partial e}{\partial x} \end{aligned} \tag{53}$$

The LTE assumption means that $T_m = T_R = T$ and the relation between the energy densities and the temperatures is given by:

$$\begin{aligned} e &= f T^\beta \rho^{-\mu} \quad \text{Rosen's notation} \\ E &= a T^4 \quad \text{Stefan-Boltzmann law} \end{aligned} \tag{54}$$

2. The finite-difference schem for this equation is: The starting point is the coupled energy equation in LTE conditions from above:

$$\begin{aligned} \frac{\partial e}{\partial t} &= \frac{\partial}{\partial x} \left(D \frac{\partial e}{\partial x} \right) \\ \frac{\partial e}{\partial t} &= \rho \frac{\partial}{\partial m} \left(D \rho \frac{\partial e}{\partial m} \right) \end{aligned} \tag{55}$$

In the Lagrangian framework used here, the cell centers are indexed by $i+\frac{1}{2}$, while the cell faces (vertices) are indexed by i. In Lagrangian coordinates, the spatial derivative is written in terms of mass coordinates, using $1/\Delta x = \rho^*/\Delta m$, where Δm is the cell mass and ρ^* is the (hydro-updated) density at the cell center.

The finite-difference (implicit) scheme for the diffusion equation in Lagrangian coordinates reads:

$$\begin{aligned} LHS &= \frac{e^{n+1}_{i+\frac{1}{2}} - e^{n}_{i+\frac{1}{2}}}{\Delta t} \\ RHS &= \frac{1}{\Delta m_{i+\frac{1}{2}}} \left[D_{i+1}\rho^*_{i+1} \frac{e^{n+1}_{i+\frac{3}{2}} - e^{n+1}_{i+\frac{1}{2}}}{\Delta m_{i+1}} - D_i \rho^*_i \frac{e^{n+1}_{i+\frac{1}{2}} - e^{n+1}_{i-\frac{1}{2}}}{\Delta m_i} \right] \end{aligned} \tag{56}$$

where D_i is the diffusion coefficient at face (vertex) i, Δm_i is the mass of the cell between vertices $i-1$ and i, and $\rho^*_{i+\frac{1}{2}}$ is the density at the cell center after the hydro update.

The harmonic mean is still used for D_i:

$$D_i = \frac{2 D_{i-\frac{1}{2}} D_{i+\frac{1}{2}}}{D_{i-\frac{1}{2}} + D_{i+\frac{1}{2}}} \tag{57}$$

(harmonic mean), and similarly for D_{i+1}.

Rearranging terms to bring all unknowns $e^{n+1}_{j+\frac{1}{2}}$ to the left yields the tridiagonal system:

$$\boxed{a_{i+\frac{1}{2}}\, e^{n+1}_{i-\frac{1}{2}} + b_{i+\frac{1}{2}}\, e^{n+1}_{i+\frac{1}{2}} + c_{i+\frac{1}{2}}\, e^{n+1}_{i+\frac{3}{2}} = e^{n}_{i+\frac{1}{2}}} \tag{58}$$

where the boxed coefficients are:

$$\boxed{a_{i+\frac{1}{2}} = -\frac{\Delta t}{\Delta m_i} \frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}} D_i} \tag{59}$$

$$\boxed{b_{i+\frac{1}{2}} = 1 + \frac{\Delta t}{\Delta m_{i+\frac{1}{2}}} \frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}} (D_i + D_{i+1})} \tag{60}$$

$$\boxed{c_{i+\frac{1}{2}} = -\frac{\Delta t}{\Delta m_{i+1}} \frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}} D_{i+1}} \tag{61}$$

Here, the index $i + \frac{1}{2}$ denotes the cell center, the left neighbor is at $i - \frac{1}{2}$ and the right neighbor at $i + \frac{3}{2}$, consistent with the Lagrangian grid convention.

For practical implementation, let $j = i + \frac{1}{2}$ and define arrays for a_j, b_j, c_j, e_j, etc. The resulting tridiagonal system is:

$$a_j\, e_{j-1}^{n+1} + b_j\, e_j^{n+1} + c_j\, e_{j+1}^{n+1} = e_j^n \tag{62}$$

which can be efficiently solved using the Thomas algorithm or any suitable tridiagonal matrix solver.

Here, $e_{i+\frac{1}{2}}^{n+1}$ is the energy at cell center $i + \frac{1}{2}$ at the new timestep, and the resulting system is tridiagonal and can be solved by the Thomas algorithm.

Note: If e and E are directly connected via LTE (e.g. $e = fT^\beta \rho^{-\mu}$ and $E = aT^4$) and the same equation holds for E, a similar discrete scheme applies to E.

5 Appendix: Derivation of the tridiagonal form coefficients in Lagrangian coordinates

1. The discretized equation for the radiation energy in Lagrangian coordinates is:

$$\frac{E_{i+\frac{1}{2}}^{n+1} - E_{i+\frac{1}{2}}^{n}}{\Delta t} + \rho_{i+\frac{1}{2}}^{*} \frac{F_{i+1}^{n+1} - F_i^{n+1}}{m_{i+\frac{1}{2}}} = \chi c \sigma_{i+\frac{1}{2}}^{n} ([U_R]_{i+\frac{1}{2}}^{n+1} - E_{i+\frac{1}{2}}^{n+1}) \tag{63}$$

2. Recalling that the flux and U_R are given as expressions of E, we can substitute them into the equation above to get a tridiagonal system for $E_{i+\frac{1}{2}}^{n+1}$.

   <u>LHS:</u> The flux is given by:

$$F_i^n = -\rho_{i+\frac{1}{2}}^{n} D_i^n \frac{E_{i+\frac{1}{2}}^{n} - E_{i-\frac{1}{2}}^{n}}{m_{i+\frac{1}{2}}} = -\rho_{i+\frac{1}{2}}^{n} \frac{2D_{i-\frac{1}{2}}^{n} D_{i+\frac{1}{2}}^{n}}{D_{i-\frac{1}{2}}^{n} + D_{i+\frac{1}{2}}^{n}} \frac{E_{i+\frac{1}{2}}^{n} - E_{i-\frac{1}{2}}^{n}}{m_{i+\frac{1}{2}}} \tag{64}$$

   and in our case (after the hydro step, and using the implicit scheme):

$$F_i^{n+1} = -\rho_{i+\frac{1}{2}}^{*} D_i^n \frac{E_{i+\frac{1}{2}}^{n+1} - E_{i-\frac{1}{2}}^{n+1}}{m_{i+\frac{1}{2}}} = -\rho_{i+\frac{1}{2}}^{*} \frac{2D_{i-\frac{1}{2}}^{n} D_{i+\frac{1}{2}}^{n}}{D_{i-\frac{1}{2}}^{n} + D_{i+\frac{1}{2}}^{n}} \frac{E_{i+\frac{1}{2}}^{n+1} - E_{i-\frac{1}{2}}^{n+1}}{m_{i+\frac{1}{2}}} \tag{65}$$

NOTE: The radiation energy density E is taken implicitely, and the D face is taken exaplicitly - because of numeric considerations (I need to ask Shay about this). Substituting the fluxes into the balance equation gives:

$$LHS = \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n}_{i+\frac{1}{2}}}{\Delta t} - \frac{\rho^{*}_{i+\frac{1}{2}}}{m_{i+\frac{1}{2}}} \overbrace{\left(\rho^{n}_{i+\frac{3}{2}} \frac{2D^{n}_{i+\frac{1}{2}} D^{n}_{i+\frac{3}{2}}}{D^{n}_{i+\frac{1}{2}} + D^{n}_{i+\frac{3}{2}}} \frac{E^{n+1}_{i+\frac{3}{2}} - E^{n+1}_{i+\frac{1}{2}}}{m_{i+\frac{3}{2}}} \right)}^{-F^{n+1}_{i+1}}$$
$$+ \frac{\rho^{*}_{i+\frac{1}{2}}}{m_{i+\frac{1}{2}}} \overbrace{\left(\rho^{n}_{i-\frac{1}{2}} \frac{2D^{n}_{i-\frac{1}{2}} D^{n}_{i+\frac{1}{2}}}{D^{n}_{i-\frac{1}{2}} + D^{n}_{i+\frac{1}{2}}} \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n+1}_{i-\frac{1}{2}}}{m_{i+\frac{1}{2}}} \right)}^{-F^{n+1}_{i}} = \tag{66}$$

Rearranging the terms gives us the tridiagonal form:

$$\begin{aligned} LHS = \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n}_{i+\frac{1}{2}}}{\Delta t} &+ \frac{\rho^{*}_{i+\frac{1}{2}}}{m^{2}_{i+\frac{1}{2}}} \left(\rho^{n}_{i+\frac{3}{2}} \frac{2D^{n}_{i+\frac{1}{2}} D^{n}_{i+\frac{3}{2}}}{D^{n}_{i+\frac{1}{2}} + D^{n}_{i+\frac{3}{2}}} \right) E^{n+1}_{i+\frac{3}{2}} \\ &- \frac{\rho^{*}_{i+\frac{1}{2}}}{m^{2}_{i+\frac{1}{2}}} \left(\rho^{n}_{i-\frac{1}{2}} \frac{2D^{n}_{i-\frac{1}{2}} D^{n}_{i+\frac{1}{2}}}{D^{n}_{i-\frac{1}{2}} + D^{n}_{i+\frac{1}{2}}} \right) E^{n+1}_{i-\frac{1}{2}} \\ &+ \frac{\rho^{*}_{i+\frac{1}{2}}}{m^{2}_{i+\frac{1}{2}}} \left(\rho^{n}_{i+\frac{3}{2}} \frac{2D^{n}_{i+\frac{1}{2}} D^{n}_{i+\frac{3}{2}}}{D^{n}_{i+\frac{1}{2}} + D^{n}_{i+\frac{3}{2}}} - \rho^{n}_{i-\frac{1}{2}} \frac{2D^{n}_{i-\frac{1}{2}} D^{n}_{i+\frac{1}{2}}}{D^{n}_{i-\frac{1}{2}} + D^{n}_{i+\frac{1}{2}}} \right) E^{n+1}_{i+\frac{1}{2}} \end{aligned} \tag{67}$$

and finally:

$$\begin{aligned} LHS = &\frac{\rho^{*}_{i+\frac{1}{2}}}{m^{2}_{i+\frac{1}{2}}} \left(\rho^{n}_{i+\frac{3}{2}} \frac{2D^{n}_{i+\frac{1}{2}} D^{n}_{i+\frac{3}{2}}}{D^{n}_{i+\frac{1}{2}} + D^{n}_{i+\frac{3}{2}}} \right) E^{n+1}_{i+\frac{3}{2}} \\ &- \frac{\rho^{*}_{i+\frac{1}{2}}}{m^{2}_{i+\frac{1}{2}}} \left(\rho^{n}_{i-\frac{1}{2}} \frac{2D^{n}_{i-\frac{1}{2}} D^{n}_{i+\frac{1}{2}}}{D^{n}_{i-\frac{1}{2}} + D^{n}_{i+\frac{1}{2}}} \right) E^{n+1}_{i-\frac{1}{2}} \\ + \left[\frac{\rho^{*}_{i+\frac{1}{2}}}{m^{2}_{i+\frac{1}{2}}} \left(\rho^{n}_{i+\frac{3}{2}} \frac{2D^{n}_{i+\frac{1}{2}} D^{n}_{i+\frac{3}{2}}}{D^{n}_{i+\frac{1}{2}} + D^{n}_{i+\frac{3}{2}}} - \rho^{n}_{i-\frac{1}{2}} \frac{2D^{n}_{i-\frac{1}{2}} D^{n}_{i+\frac{1}{2}}}{D^{n}_{i-\frac{1}{2}} + D^{n}_{i+\frac{1}{2}}} \right) + \frac{1}{\Delta t} \right] &E^{n+1}_{i+\frac{1}{2}} - \frac{1}{\Delta t} E^{n}_{i+\frac{1}{2}} \end{aligned} \tag{68}$$

<u>RHS:</u> Substituting the expression for $[U_R]^{n+1}_{i+\frac{1}{2}}$ gives:

$$
\begin{aligned}
RHS &= \chi c \sigma^n_{i+\frac{1}{2}} \left(\frac{A^n_{i+\frac{1}{2}}}{1+A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}} + \frac{1}{1+A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - E^{n+1}_{i+\frac{1}{2}} \right) \\
&= \chi c \sigma^n_{i+\frac{1}{2}} \left(\frac{1}{1+A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - \frac{1}{1+A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}} \right) \\
&= \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1+A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1+A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}}
\end{aligned}
\tag{69}
$$

3. Finally, rearranging the equation $LHS = RHS$ gives us the tridiagonal form:

$$
a^n_i E^{n+1}_{i-\frac{1}{2}} + b^n_i E^{n+1}_{i+\frac{1}{2}} + c^n_i E^{n+1}_{i+\frac{3}{2}} = d^n_i \tag{70}
$$

and the coefficients are given by:

$$
\boxed{a^n_i = -\frac{\rho^*_{i+\frac{1}{2}}}{m^2_{i+\frac{1}{2}}} \left(\rho^n_{i-\frac{1}{2}} \frac{2D^n_{i-\frac{1}{2}} D^n_{i+\frac{1}{2}}}{D^n_{i-\frac{1}{2}} + D^n_{i+\frac{1}{2}}} \right)} \tag{71}
$$

$$
\boxed{b^n_i = \frac{\rho^*_{i+\frac{1}{2}}}{m^2_{i+\frac{1}{2}}} \left(\rho^n_{i+\frac{3}{2}} \frac{2D^n_{i+\frac{1}{2}} D^n_{i+\frac{3}{2}}}{D^n_{i+\frac{1}{2}} + D^n_{i+\frac{3}{2}}} - \rho^n_{i-\frac{1}{2}} \frac{2D^n_{i-\frac{1}{2}} D^n_{i+\frac{1}{2}}}{D^n_{i-\frac{1}{2}} + D^n_{i+\frac{1}{2}}} \right) + \frac{1}{\Delta t} + \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1+A^n_{i+\frac{1}{2}}}} \tag{72}
$$

$$
\boxed{c^n_i = \frac{\rho^*_{i+\frac{1}{2}}}{m^2_{i+\frac{1}{2}}} \left(\rho^n_{i+\frac{3}{2}} \frac{2D^n_{i+\frac{1}{2}} D^n_{i+\frac{3}{2}}}{D^n_{i+\frac{1}{2}} + D^n_{i+\frac{3}{2}}} \right)} \tag{73}
$$

$$
\boxed{d^n_i = \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1+A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} + \frac{1}{\Delta t} E^n_{i+\frac{1}{2}}} \tag{74}
$$

where $A^n_{i+\frac{1}{2}} = \beta^n_{i+\frac{1}{2}} \Delta t \chi c \sigma^n_{i+\frac{1}{2}}$ is the coupling coefficient between the radiation and matter. I need to understand how the Flek & Cummings coefficient f is related to the coupling coefficient A).

5.1 Deeper understanding of the tridiagonal form

1. First, let's make order in the celebration of indexes. Our staggered grid is as follows:

$i-1$	$i-\frac{1}{2}$	i	$i+\frac{1}{2}$	$i+1$	$i+\frac{3}{2}$
$j-\frac{3}{2}$	$\boxed{j-1}$	$j-\frac{1}{2}$	$\boxed{j}$	$j+\frac{1}{2}$	$\boxed{j+1}$
node	cell center	node	cell center	node	cell center
	$E_{i-\frac{1}{2}}$		$E_{i+\frac{1}{2}}$		$E_{i+\frac{3}{2}}$
	$[U_R]_{i-\frac{1}{2}}$		$[U_R]_{i+\frac{1}{2}}$		$[U_R]_{i+\frac{3}{2}}$
	$\rho_{i-\frac{1}{2}}$		$\rho_{i+\frac{1}{2}}$		$\rho_{i+\frac{3}{2}}$
	$D_{i-\frac{1}{2}}$		$D_{i+\frac{1}{2}}$		$D_{i+\frac{3}{2}}$
	$\beta_{i-\frac{1}{2}}$		$\beta_{i+\frac{1}{2}}$		$\beta_{i+\frac{3}{2}}$
	$\sigma_{i-\frac{1}{2}}$		$\sigma_{i+\frac{1}{2}}$		$\sigma_{i+\frac{3}{2}}$

where we defined $j = i + \frac{1}{2}$ as the index for the cell centers. Now, the indexes values are:

$$\begin{aligned} i &= 0, 1, 2, ..., N \quad \text{(nodes)} \\ j &= 1, 2, ..., N \quad \text{(cell centers)} \end{aligned} \tag{75}$$

2. We wish to write the tridiagonal equation as:

$$a_j^n E_{j-1}^{n+1} + b_j^n E_j^{n+1} + c_j^n E_{j+1}^{n+1} = d_j^n \tag{76}$$

Before we think of the definitions for the coefficients, let's understand how this equation can be written in matrix form. Supposing that E_0 is given as a BC (discussed later) our unknowns are $E_1, E_2, ..., E_N$ (remember the E is indexed by j, the cell centers). The boundary conditions for a power-law self-similar solution with zero radiation energy at the end of the grid are:

$$E_0^{n+1} = a_{\text{Hev}} T_0 t^\tau \tag{77}$$

$$E_{N-1}^{n+1} = 0 \tag{78}$$

Two notes: (1) This derivation holds for any BCs, even without a BC at the end of the grid that will give us an equation for E_{N-1} and make $a_{N-1}, b_{N-1}, c_{N-1}, d_{N-1}$ non-zero. (2) I need to ask Shay about the correct BC here and its physical meaning.

Therefore, the entire matrix equation for the vector of unknowns is:

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ a_2 & b_2 & c_2 & \cdots & 0 & 0 \\ 0 & a_3 & b_3 & c_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{N-2} & b_{N-2} & c_{N-2} \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} E_1^{n+1} \\ E_2^{n+1} \\ E_3^{n+1} \\ \vdots \\ E_{N-1}^{n+1} \\ E_N^{n+1} \end{bmatrix} = \begin{bmatrix} a_{\text{Hev}} T_0 t^\tau \\ d_2 \\ d_3 \\ \vdots \\ d_{N-2} \\ 0 \end{bmatrix} \tag{79}$$

From now, we can reduce the a_1 and the c_{N-2} from the 0th and $(N-2)^{th}$ row and get the matrix equation:

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & b_2 & c_2 & \cdots & 0 & 0 \\ 0 & a_3 & b_3 & c_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{N-2} & b_{N-2} & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} E_1^{n+1} \\ E_2^{n+1} \\ E_3^{n+1} \\ \vdots \\ E_{N-1}^{n+1} \\ E_N^{n+1} \end{bmatrix} = \begin{bmatrix} a_{\text{Hev}} T_0 t^\tau \\ d_2 - a_{\text{Hev}} T_0 t^\tau \\ d_3 \\ \vdots \\ d_{N-2} \\ 0 \end{bmatrix} \tag{80}$$

Notice that we thereby received an invariant matrix equation of $N-2$ equations in $N-2$ unknowns, which is:

$$\begin{bmatrix} b_2 & c_2 & \cdots & 0 & 0 \\ a_3 & b_3 & c_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{N-2} & b_{N-2} \end{bmatrix} \begin{bmatrix} E_2^{n+1} \\ E_3^{n+1} \\ \vdots \\ E_{N-1}^{n+1} \end{bmatrix} = \begin{bmatrix} d_2 - a_{\text{Hev}} T_0 t^\tau \\ d_3 \\ \vdots \\ d_{N-2} \end{bmatrix} \tag{81}$$

This is a *tridiagonal system of equations* which can be solved by the *Thomas algorithm* which is explained in previous files.

3. Now, we can define the coefficients $a_j^n, b_j^n, c_j^n, d_j^n$ for $j = 2, ..., N-2$ (the index $N-1$ is irrelevant for the end BC that we have, but it will be relevant if we have a different BC at the end of the grid). The transition is simply $j = i + \frac{1}{2}$, so we can rewrite the coefficients in terms of j instead of i:

$$\begin{aligned} a_j^n &= -\frac{\rho_j^*}{m_j^2}\left(\rho_{j-1}^n \frac{2D_{j-1}^n D_j^n}{D_{j-1}^n + D_j^n}\right) \\ b_j^n &= \frac{\rho_j^*}{m_j^2}\left(\rho_{j+1}^n \frac{2D_j^n D_{j+1}^n}{D_j^n + D_{j+1}^n} - \rho_{j-1}^n \frac{2D_{j-1}^n D_j^n}{D_{j-1}^n + D_j^n}\right) + \frac{1}{\Delta t} + \frac{\chi c \sigma_j^n}{1 + A_j^n} \\ c_j^n &= \frac{\rho_j^*}{m_j^2}\left(\rho_{j+1}^n \frac{2D_j^n D_{j+1}^n}{D_j^n + D_{j+1}^n}\right) \\ d_j^n &= \frac{\chi c \sigma_j^n}{1 + A_j^n}[U_R]_j^n + \frac{1}{\Delta t} E_j^n \end{aligned} \tag{82}$$

4. IMPORTANT NOTE: ρ, m, D, σ are all given as arrays for the cell centers (i.e. $\rho_j =$ `rho[j - 1]`) for $j = 1, ..., N$. However, even for

the non-zero BC at $j = N$, the coefficients $a_j^n, b_j^n, c_j^n, d_j^n$ are defined for $j = 2, ..., N-1$ so we will define them so that $a_j =$ `a[j - 2]` (so that `a.size()=N-2` and `a=[a[0],a[1],...,a[N-3]]`). Therefore, each time a variable ρ_j appears in the formula for a_j (for example), we will replace a_j by `a[j-2]` and ρ_j by `rho[j-1]`. **To sum up**, the arrays look like this:

```
a=[a[0],a[1],...,a[N-3]]
rho=[rho[0],rho[1],...,rho[N-1]]
```
(83)

and the rules for translating the formulas for the coefficients into array assignment with the correct indexing, are:

$$\begin{aligned} a_j^n &\to \texttt{a[:]} \quad \texttt{or just a} \\ \rho_j^n &\to \texttt{rho[1:-1]} \\ \rho_{j-1}^n &\to \texttt{rho[:-2]} \\ \rho_{j+1}^n &\to \texttt{rho[2:]} \end{aligned} \tag{84}$$

and, of course, the same for the other coefficients & variables.